

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Naveen Kumar (192521412) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1: Smart Home Automation System Using Intelligent Agents

Introduction

A smart home automation system manages lighting, temperature, and security using sensor inputs such as motion detection, temperature changes, and time of day. It also learns user preferences to improve comfort, energy efficiency, and safety.

Types of Intelligent Agents

1. Simple Reflex Agent

Acts only on the current sensor input using condition-action rules. Example: Turn on lights when motion is detected at night.

2. Model-Based Reflex Agent

Maintains an internal model of the environment. Example: Remembers whether lights are already on before taking action.

3. Goal-Based Agent

Selects actions that help achieve goals such as maintaining a comfortable room temperature while minimizing power consumption.

4. Utility-Based Agent

Chooses the action with the highest overall benefit by balancing comfort, energy savings, and security.

Analysis

Simple Reflex Agents are fast but cannot learn. Model-Based Agents handle incomplete information. Goal-Based Agents make better long-term decisions. Utility-Based Agents optimize multiple objectives such as comfort, cost, and safety.

Most Effective Approach

A Hybrid Intelligent Agent is the best choice because it combines reflex actions for immediate responses, model-based reasoning for context awareness, goal-based planning for automation, and utility-based decision-making for optimal comfort, energy efficiency, and security.

Conclusion

Hybrid agents provide the most reliable smart home system because they react quickly, learn user preferences, adapt to changing conditions, and optimize overall performance.

Question 2: Robot Navigation in an Unknown Maze

Problem Statement

A robot is placed in an unknown maze and must find an exit without prior knowledge of the layout.

Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the lowest path cost first. It guarantees the least-cost solution when step costs vary but may require significant memory and computation.

Iterative Deepening Search (IDS)

Iterative Deepening Search repeatedly performs depth-limited searches with increasing limits. It combines the low memory usage of DFS with the completeness of BFS when all step costs are equal.

Comparison

Feature	UCS	IDS
Completeness	Yes	Yes
Optimality	Yes (least cost)	Yes (equal costs)
Memory	High	Low
Time	Can be slow	Repeated searches increase time

Relation to Intelligent Agents

A Goal-Based Agent is most suitable because it searches for actions that achieve the objective of reaching the exit. The search strategy determines the path, while the agent decides which actions to perform.

Recommended Combination

A Goal-Based Agent with Uniform Cost Search is the most effective when movement costs differ because it guarantees the lowest-cost path. If all movements have equal cost and memory is limited, Goal-Based Agent with Iterative Deepening Search is an excellent alternative.

Overall Conclusion

Different intelligent agents are designed for different environments. Hybrid agents are ideal for smart homes, while Goal-Based agents combined with appropriate search strategies enable robots to solve navigation problems efficiently.